TI TIDA-060029 analog front end

Matched RG/RF: 100R / 5k / 100k

File: afe.kicad_sch

Codec DAC excitation

Coherent codec DAC excitation

File: source.kicad_sch

Codec acquisition (TAC5112)

Two channels / one frame clock

File: codec.kicad_sch

Range relays

Relay-selected ranges

File: range.kicad_sch

USB, RP2040 and boot / debug

USB control, firmware and host calibration

File: digital.kicad_sch

Power: 5V input, forced PWM, quiet analog rails

External 5V → quiet +/-12V

File: power.kicad_sch

Engineering prototype. One PCB. Isolated, discharged DUT only.

**Temper**

Sheet: /
File: temper-lcr.kicad_sch

***Title: Temper LCR — TIDA-060029 based***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-20 | **Rev: E** |
| KiCad E.D.A. 10.0.4 | | Id: 7/7 |

U6A source buffer; U6B HI sense; U7A transimpedance; U7B LO sense.
Ranges: latching relays K1/K2/K3 (range sheet) close one matched RG/RF pair.
One local C104=100pF compensates all ranges; C78 100pF DUT_LO to GND (Rev E stability).
No BUF634A: maximum source current approximately 18mA. Isolated, discharged DUT only.

## TI source and range network

U6A
EXCITE
OUT
OPA2810IDGKR
SOURCE_BUF

R40
RG100
100R 0.1%
R41
RG5K
5k 0.1%
R42
RG100K
100k 0.1%

J3
DUT_HI
DUT_LO
DUT HI / LO

J3: connect external, isolated DUT between HI and LO.
No DUT component is fitted on this PCB.

C78
DUT_LO
GND
100pF C0G 5%
Rev E: C78 100pF C0G from J3 LO to GND (TIA margin with relay parasitics).

## TI transimpedance topology; Rev C compensation
C104=100pF directly across U7.1 and U7.2 in all ranges

Rev D: R94 22R 0.1% between J3 LO and the TIA summing node.
LOW follower senses J3 LO; host uses Zf = R94 + RF||C104.

R94
22R 0.1%
TIA_SUM

R43
RF100
100R 0.1%
R44
RF5K
5k 0.1%
R45
RF100K
100k 0.1%

C104
100pF C0G 5%

U7A
GND
OUT
OPA2810IDGKR
TIA_OUT

U6B
HIGH_BUF
OPA2810IDGKR

U7B
LOW_BUF
OPA2810IDGKR

## Independent HI / LO voltage followers

U6C
12V
-12V
V+
V-
OPA2810IDGKR

C42
100nF 50V
C43
100nF 50V
C44
1uF 50V
C45
1uF 50V
C50
10nF 50V
C51
10nF 50V

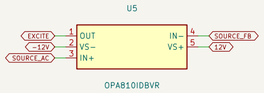
U7C
OPA2810IDGKR

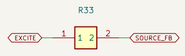
C46
100nF 50V
C47
100nF 50V
C48
1uF 50V
C49
1uF 50V
C52
10nF 50V
C53
10nF 50V

Retain TI 100nF + 1uF bypass per rail; added 10nF at each supply pin.
PCB plane windows and short feedback routes are specified in reference-conformance.md.

| | |
|---|---|
| Temper | |
| Sheet: /TI TIDA-060029 analog front end/ | |
| File: afe.kicad_sch | |
| **Title: TI TIDA-060029 analog front end** | |
| Size: A2 | Date: 2026-10-20 |
| KiCad E.D.A. 10.0.4 | **Rev: E** |
| | Id: 2/7 |

TAC5112 OUT1P/OUT1M differential line out at fs = 187.5kHz.
OPA810 difference stage: 2.2k in, 2.0k || 270pF feedback and ground leg, gain 0.909.
DC coupled: the stage rejects the 1.65V DAC common mode.





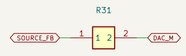


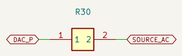


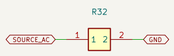













Global net names connect functional sheets. See TI conformance matrix.

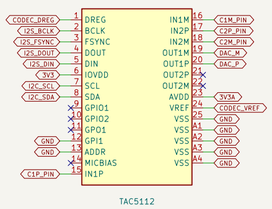
**Temper**

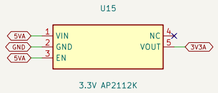
Sheet: /Codec DAC excitation/
File: source.kicad_sch

***Title: Codec DAC excitation***

| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-20 | **Rev: E** |
| KiCad E.D.A. 10.0.4 | | Id: 3/7 |

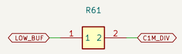
CH1 = HIGH_BUF – LOW_BUF (IN1P/IN1M); CH2 = LOW_BUF – TIA_OUT (IN2P/IN2M).
Each leg: 1k series, 1k + 560pF to GND (K = 0.5), 1uF, BAT54S clamp, 470R to the pin.
TDM controller: RP2040 drives BCLK 12MHz / FSYNC 187.5kHz. I2C address 0x50.
AVDD from U15 AP2112K-3.3 on 5VA; IOVDD from 3V3.

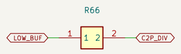


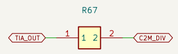



























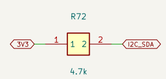


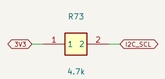


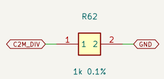








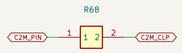


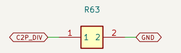










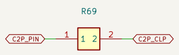


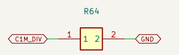








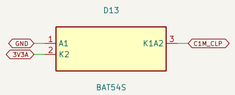


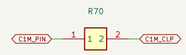



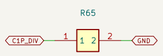






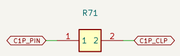













Global net names connect functional sheets. See TI conformance matrix.

**Temper**

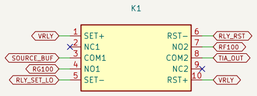
Sheet: /Codec acquisition (TAC5112)/
File: codec.kicad_sch

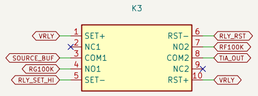
***Title: Codec acquisition (TAC5112)***

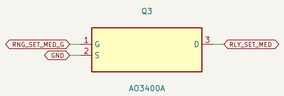
| | | |
|---|---|---|
| Size: A2 | Date: 2026-10-20 | **Rev: E** |
| KiCad E.D.A. 10.0.4 | | Id: 4/7 |

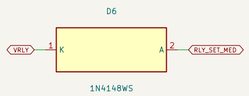
K1 100R, K2 5k, K3 100k: Panasonic TQ2SA-L2-5V dual-coil latching DPDT.
Pole 1: SOURCE_BUF -> RGx. Pole 2: TIA_OUT -> RFx. NC contacts unconnected.
RESET coils in parallel on Q2; SET coils on Q3/Q4/Q5. 1N4148WS flyback per drain.
VRLY = 5VA through 2.2R + 22uF. Firmware: source off, RESET 20ms, SET 20ms.

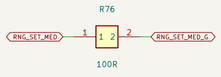


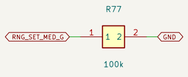



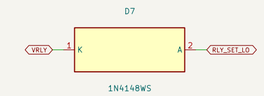



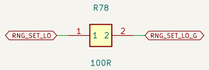



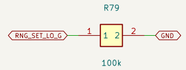



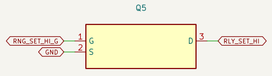






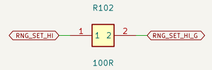



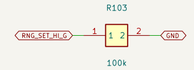




























Global net names connect functional sheets. See TI conformance matrix.

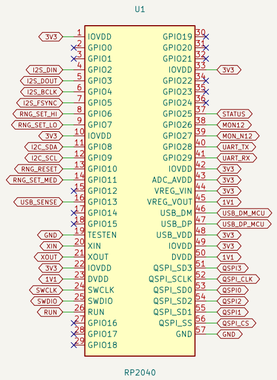
**Temper**

Sheet: /Range relays/
File: range.kicad_sch

***Title: Range relays***

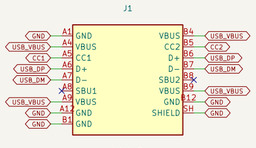
| Size: A2 | Date: 2026-10-20 | **Rev: E** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 5/7 |

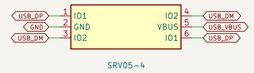
External 5V input powers the instrument. USB VBUS is sensed only; no backfeed.
Native USB CDC; 120MHz MCU; dual-lane ADC acquisition at up to 1MSPS.
Hold BOOTSEL while resetting for ROM USB firmware loading. SWD and UART provided.





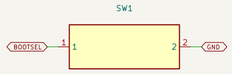


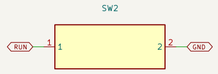


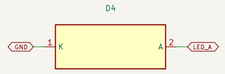



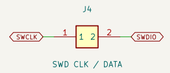


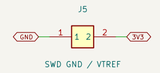


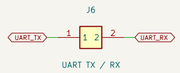


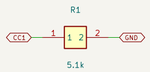



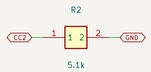



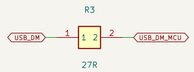



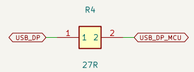



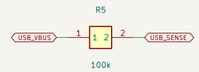











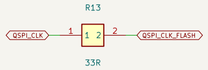






















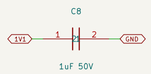














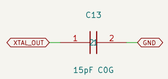






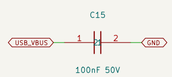


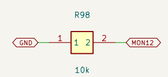


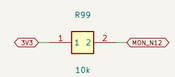
















Global net names connect functional sheets. See TI conformance matrix.

| | |
|---|---|
| **Temper** | |
| Sheet: /USB, RP2040 and boot / debug/ | |
| File: digital.kicad_sch | |
| ***Title: USB, RP2040 and boot / debug*** | |
| Size: A2 | Date: 2026-10-20 |
| KiCad E.D.A. 10.0.4 | **Rev: E** |
| | Id: 6/7 |

Input: REGULATED 5V +/−5%, 1A, center positive. USB does not supply this board.
TPS65131 PSP/PSN grounded: forced PWM 1.25−1.5MHz. RAW_P −14.31V: RAW_N −−14.39V.
TPS7A49 −+11.97V: TPS7A30 −−12.00V. Fixed output values are nominal.
Do not substitute capacitor voltage/package without checking DC−bias derating.

U3
3.3V AP2112K

U11
TPS7A4901DGNR

U12
TPS7A3001DGNR

U13
TPS65131RGER

Q1
AO3401A

D2
SS14

D3
SS14

L1
4.7uH 3.8A shielded

L2
4.7uH 3.8A shielded

FB1
100R@100MHz 800mA

J2
5V 1A CENTER+

R95
1R

R96
1R

R80
100R

R81
1.62M

R82
150k

R83
1.74M

R84
150k

R85
91k 0.1%

R86
10k 0.1%

R87
91k 0.1%

R88
1k 0.1%

R89
10k 0.1%

C16
1uF 50V

C17
1uF 50V

C18
10uF 25V X5R

C80
10uF 50V X7R

C81
100nF 50V

C82
10uF 50V X7R

C83
100nF 50V

C84
220nF 25V

C85
4.7nF 50V

C86
10nF 50V

C87
4.3pF C0G

C88
4.3pF C0G

C89
10uF 50V X7R

C90
10uF 50V X7R

C91
10uF 50V X7R

C92
10uF 50V X7R

C93
10nF 50V

C94
10uF 50V X7R

C95
10uF 50V X7R

C96
10nF 50V

C97
100nF 50V

C98
10nF 50V

C99
10uF 50V X7R

C100
10uF 50V X7R

C101
10nF 50V

C102
100nF 50V

C103
22uF 25V X5R

R97
100k

R100
150k

PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG

Global net names connect functional sheets. See TI conformance matrix.

**Temper**

Sheet: /Power: 5V input, forced PWM, quiet analog rails/
File: power.kicad_sch

***Title: Power: 5V input, forced PWM, quiet analog rails***

| | | |
|---|---|---|
| Size: A2 | Date: 2026−10−20 | **Rev: E** |
| KiCad E.D.A. 10.0.4 | | Id: 7/7 |